

Lab Transaction ACID Properties

Exercise 1: Testing the "All or Nothing"

Initial check

Query

Query History

1

2

SELECT rental_id, customer_id FROM rental;

Data Output

Messages

Notifications

</

Intentional Error


```
Query  Query History
1  BEGIN;
2
3  INSERT INTO customer (first_name, last_name, email, address_id, store_id)
4  VALUES ('AI', 'Student', 'ai@edu.vn', 1, 1);
5
6  SAVEPOINT user_created;
7
8  INSERT INTO special_offers (customer_id, code) VALUES (9999, 'FREE');
```

Data Output Messages Notifications

```
ERROR:  relation "special_offers" does not exist
LINE 8: INSERT INTO special_offers (customer_id, code) VALUES (9999,...
                  ^

SQL state: 42P01
Character: 166
```

Partial rollback

```
Query  Query History
1      ROLLBACK TO user_created;
2
3      COMMIT;

Data Output  Messages  Notifications
COMMIT

Query returned successfully in 84 msec.
```

Check the result

Query

Query History

1

```
SELECT first_name, last_name, email FROM customer WHERE email = 'ai@edu.vn';
```

Data Output

Messages

Notifications

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SQL

Showing

	first_name character varying (45) 🔒	last_name character varying (45) 🔒	email character varying (50) 🔒
1	AI	Student	ai@edu.vn

Exercise 2: The "Flexible" Staff Task

Initial setup and Intentional error

```
Query  Query History
1  BEGIN;
2
3  -- 1. Đăng ký một nhân viên mới (Gán tạm vào store 1)
4  INSERT INTO staff (first_name, last_name, address_id, store_id, username)
5  VALUES ('Data', 'Scientist', 1, 1, 'datascientist');
6
7  -- 2. Tạo điểm lưu
8  SAVEPOINT staff_created;
9
10 -- 3. Cố ý tạo lỗi: Gán nhân viên vào chi nhánh không tồn tại
11 UPDATE staff SET store_id = 'LỖI_KIẾU_DỮ_LIỆU' WHERE username = 'datascientist';
12
13

Data Output  Messages  Notifications
ERROR:  invalid input syntax for type smallint: "LỖI_KIẾU_DỮ_LIỆU"
LINE 11: UPDATE staff SET store_id = 'LỖI_KIẾU_DỮ_LIỆU' WHERE usernam...
                                   ^
SQL state: 22P02
Character: 326
```

Partial rollbacks to staff_created and commit

```
Query  Query History
1  ROLLBACK TO staff_created;
2
3  UPDATE staff SET store_id = 1 WHERE username = 'datascientist';
4
5  COMMIT;
6
7

Data Output  Messages  Notifications
COMMIT
Query returned successfully in 84 msec.
```

Check the results

Query Query History

```
1 SELECT first_name, last_name, store_id, username
2 FROM staff
3 WHERE username = 'datascientist';
4
```

Data Output Messages Notifications



SQL

first_name
character varying (45) 🔒

last_name
character varying (45) 🔒

store_id
smallint 🔒

username
character varying (16) 🔒

1

Data

Scientist

1

datascientist